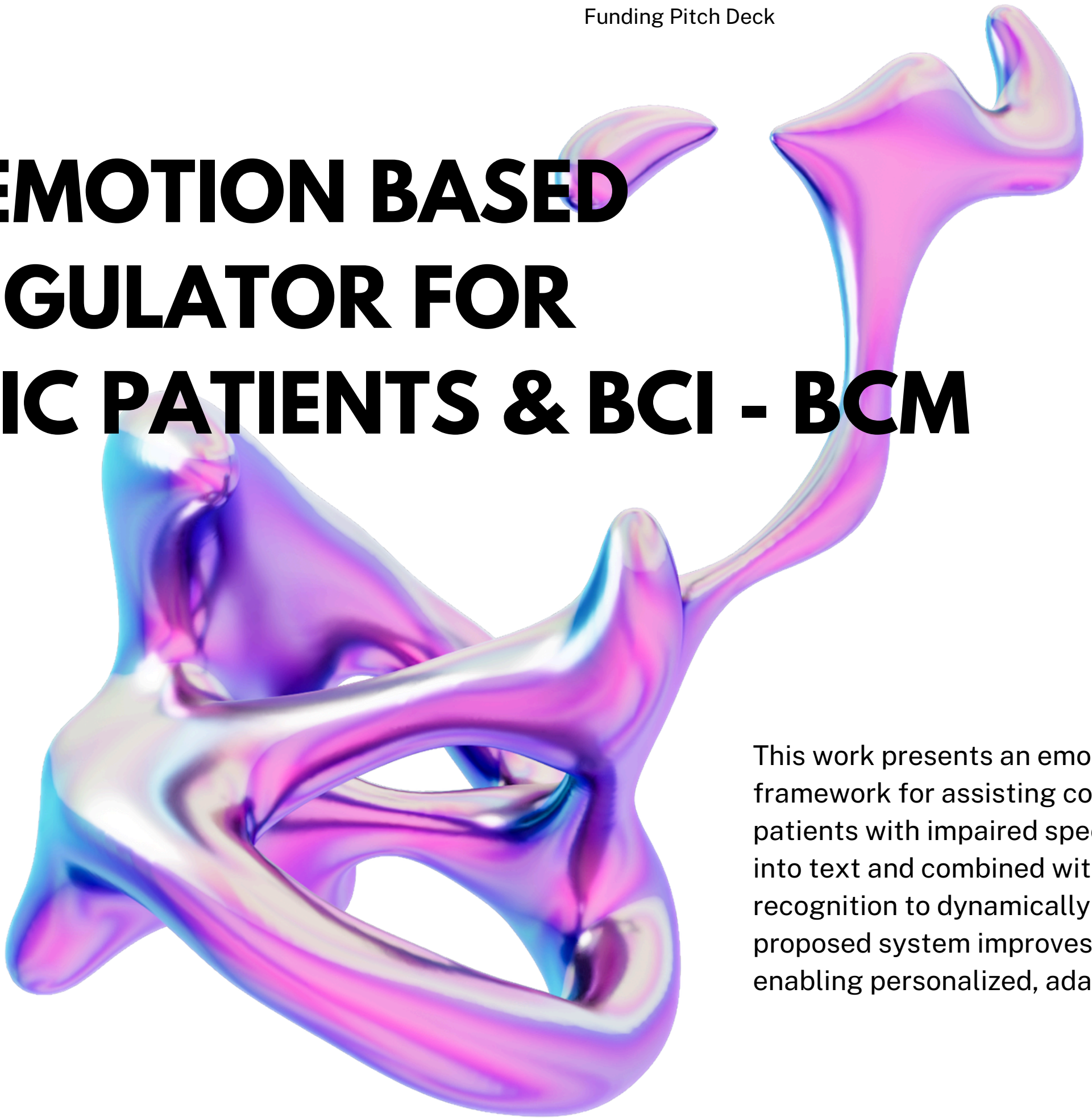


# AI DRIVEN EMOTION BASED DOSAGE REGULATOR FOR PSYCHIARTIC PATIENTS & BCI - BCM



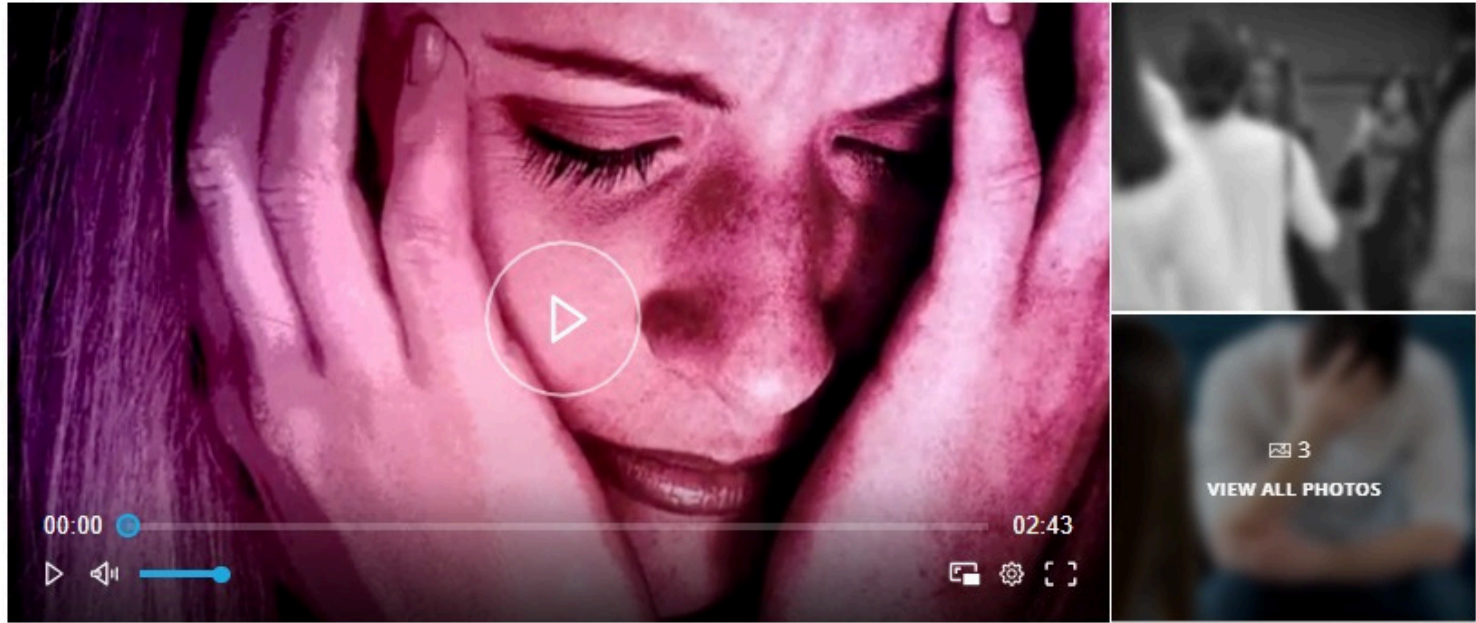
This work presents an emotion-aware brain-to-text decoding framework for assisting communication in psychiatric patients with impaired speech. Neural signals are decoded into text and combined with real-time emotional state recognition to dynamically regulate medication dosage. The proposed system improves clinical responsiveness while enabling personalized, adaptive mental health care.

# RESEARCH & STUDY

01

## Suicide rates for 2023 at an all-time high for the United States

by Karris Harmon | Thu, January 4th 2024 at 5:24 PM  
Updated Thu, January 4th 2024 at 6:15 PM



Suicide rates for 2023 at an all-time high for the United States (WPMI)



TOPICS: [SUICIDE](#) [MENTAL HEALTH](#) [RATES](#) [PROFESSIONALS](#) [INCREASE](#) [CRISIS](#) [HOPE](#) [ECONOMY](#)

MOBILE, Ala. (WPMI) — In 2022, the CDC said more than 49,000 people in the United States died by suicide. In 2023, those numbers surpassed 50,000 suicide deaths. Mental health officials say it is the highest rate of suicide that the nation has ever seen. Local mental health professionals I spoke with say the numbers are alarming, but they are hopeful that things will get better.

02



## Overdose Deaths Drop More Than 25%, Continuing Trend

Naloxone distribution, increased access to buprenorphine, and changes in opioid prescribing patterns have spurred a rapid decline in overdose deaths. But at a time of threatened state funding, will that trend continue? BY TERRI D'ARRIGO

Annual overdose deaths declined more than 25% nationwide for the 12 months ending October 31, 2024—from 109,703 to 82,020—compared with the same time period one year earlier, according to

Statistics (NCHS). The drop continues a trend of decreasing overdose deaths that began early in the fall of 2023. "These numbers mean that there are more than 75 fewer overdose deaths per day," Allison Arwady, M.D., M.P.H.,

increased overdose deaths during the height of the pandemic, likely from disruptions in treatment and an increase in people using substances while they were alone, this decrease in overdose deaths is likely not just a result of coming out of the pandemic."

### A Tale of Two States

North Carolina had the greatest decline in overdose deaths, from 3,852

the educational requirements from licensees. They were getting tired of seeing patients struggling with addiction and putting their own medical licenses on the line. People began to realize the secondary consequences of just prescribing [opioids] as they wanted."

Medical education has also evolved in the state. "We talk about this in our medical schools, and we're no longer

# RESEARCH & STUDY

03

Health

**Karnataka has highest suicide-related economic burden in country: Report**

*The southern state alone accounts for over a fifth of the national economic losses due to suicides*



Family members of a chilli farmer in Telangana who died by suicide after the black thrips pest destroyed his entire crop in 2022. Archival photo for representation: Vikas Choudhary / CSE

Coovercolly Indresh

Published on: 19 Sep 2024, 2:14 pm

India is grappling with a suicide crisis, which has been found to carry a significant economic burden. The country loses some Rs 13.4 lakh crore (\$16 billion) due to suicides every year, according to recent study published in *The Lancet*.

The southern state of Karnataka, with only 5 per cent of the country's population, shoulders the heaviest economic burden from suicides, contributing a little over a fifth of the national total, or ₹2.33 lakh crore.

04

RAJASTHAN ELECTIONS PHOTOS WEATHER

NEWS / CITY NEWS / JAIPUR NEWS / Drug Overdose Likely Cause Of Death Of 2 Bikaner Men

**TRENDING** Siddharth Yadav Death Delhi Airport Marijuana Seized Hyderabad Murder Bijnor Murder Sanjana Galran

## Drug overdose likely cause of death of 2 Bikaner men

TNN / Mar 31, 2025, 04:44 IST

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Jaipur: A day after two decomposed bodies were recovered from a house in Bikaner town, the police said Sunday that one of the deceased was working as a male nurse at the govt-run PBM Hospital in Bikaner, while the other was his friend. Both are suspected to have died of a drug overdose as several liquor bottles and syringes were found lying in the house. The bodies were decomposed beyond identification, and going by

their condition, the police suspect that they might have been lying in the house for the past six to seven days.

Police later identified the deceased as Dharmendra Tanvar, a nursing staff member at PBM Hospital, and his friend, Pramod Kumar Sharma. Pramod's body was found lying on a sofa, while Dharmendra's was on the floor.

"Syringes and empty liquor bottles were scattered across the room, following which we suspect that

# EXISTING PROBLEMS



## **UNPREDICTABLE MOOD SWINGS**

*Bipolar patients often face sudden, severe emotional shifts that are hard to track manually.*



## **INCONSISTENT MONITORING**

*Traditional methods are subjective and irregular, relying on therapist observations or patient self-reports.*



## **LACK OF AI TOOLS**

*There is no system that can continuously and objectively monitor emotional and behavioral changes in real-time.*



## **ONE-SIZE-FITS-ALL DOSAGE**

*Most treatments follow a fixed dosage model, ignoring the patient's day-to-day mental state.*

**We Propose - An AI Driven Dosage Regulator for Psychiatric patients & BCI**

**1) Real-Time Emotion Detection Using Multimodal AI Facial expressions, speech patterns, and behavioral cues are continuously analyzed using deep learning models to identify emotional states such as anxiety, depression, and agitation with high accuracy.**

**2) Brain-Computer Interface (BCI) for Direct Neural Emotion Sensing Neural signals related to cognitive load and emotional stress are captured through non-invasive BCI devices, enabling detection of internal emotional changes even when external expressions are minimal or suppressed.**

# Solution

An AI Driven Dosage Regulator for Psychiatric patients & BCI



## **EMOTION MONITORING VIA AI**

*Uses camera and mic to analyze facial expressions, voice tone, and behavior – providing continuous emotional state assessment.*



## **DYNAMIC DOSAGE RECOMMENDATION**

*An AI/ML model maps real-time + historical mood data to suggest personalized medication doses, adapting to daily mental changes.*



## **SMART PILL DISPENSING SYSTEM**

*Integrates an ESP32-based dispenser that auto-administers doses per AI guidance, with manual override and safety checks.*



## **CAREGIVER DASHBOARD**

*A secure web portal displays emotional trends, dosage logs, and alerts – enabling caregivers and doctors to review and adjust treatment.*

**3. Emotion-Health Mapping Engine for Psychiatric State Assessment** AI models fuse emotion data and BCI signals to quantify mental health severity levels, creating a real-time psychiatric condition index for each patient.

**4. Adaptive Medication Dosage Regulation System** Based on the assessed emotional state, an intelligent dosage controller dynamically adjusts medication recommendations to prevent relapse, overdose, or ineffective treatment.

## So , Where does BCI Comes into part ?

# ANALYSIS DONE

## VOICE TONE RECOGNITION



*Analyzes speech patterns to detect emotional cues such as stress, anxiety, or depression.*

## FACIAL EXPRESSION RECOGNITION



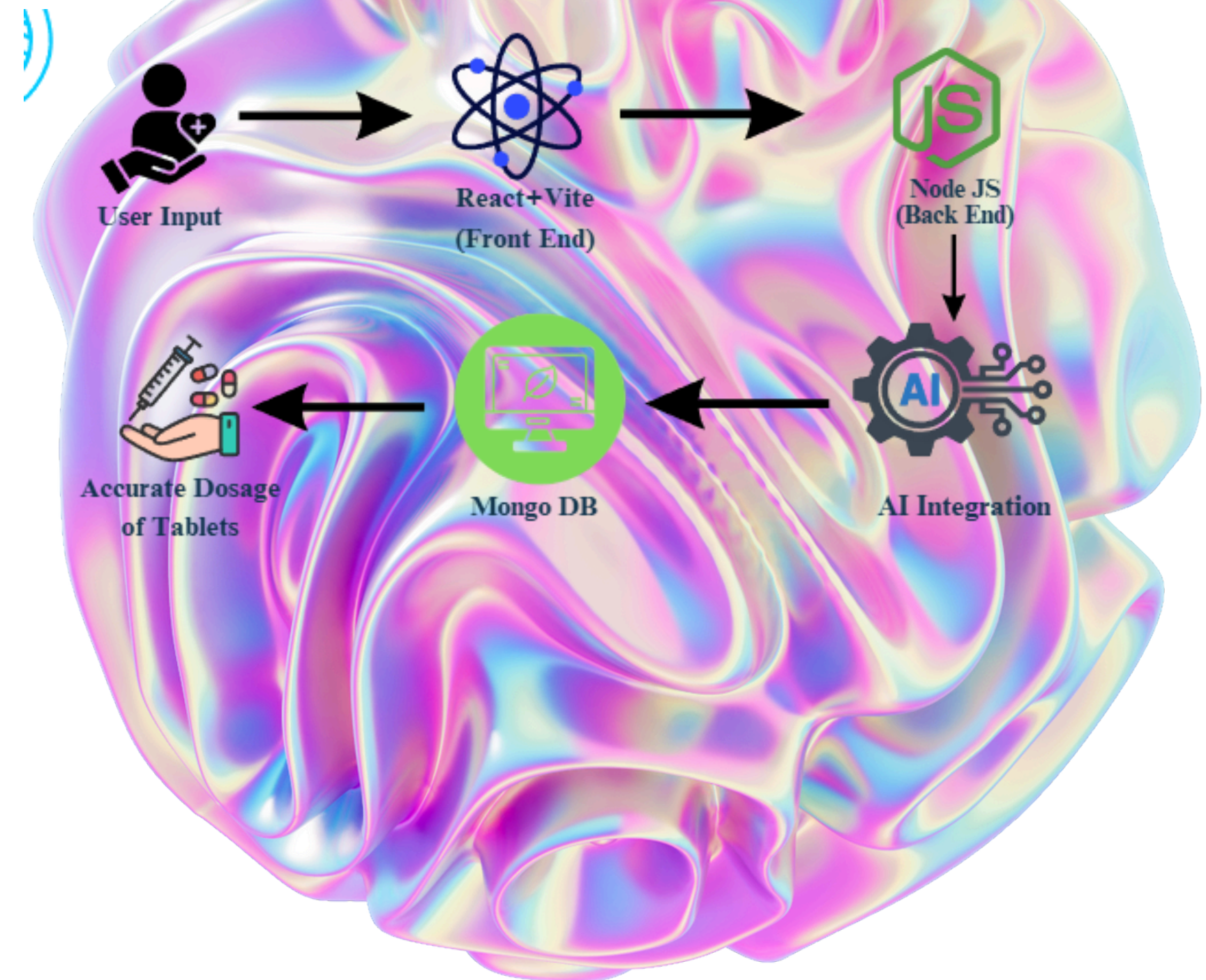
*Assesses facial expressions to identify emotions like sadness, agitation, or calmness.*

## COMBINED RECOGNITION



*Integrates both voice and facial data for a more accurate emotional assessment.*

# TECH STACKS



# BRAIN TO TEXT - BCI

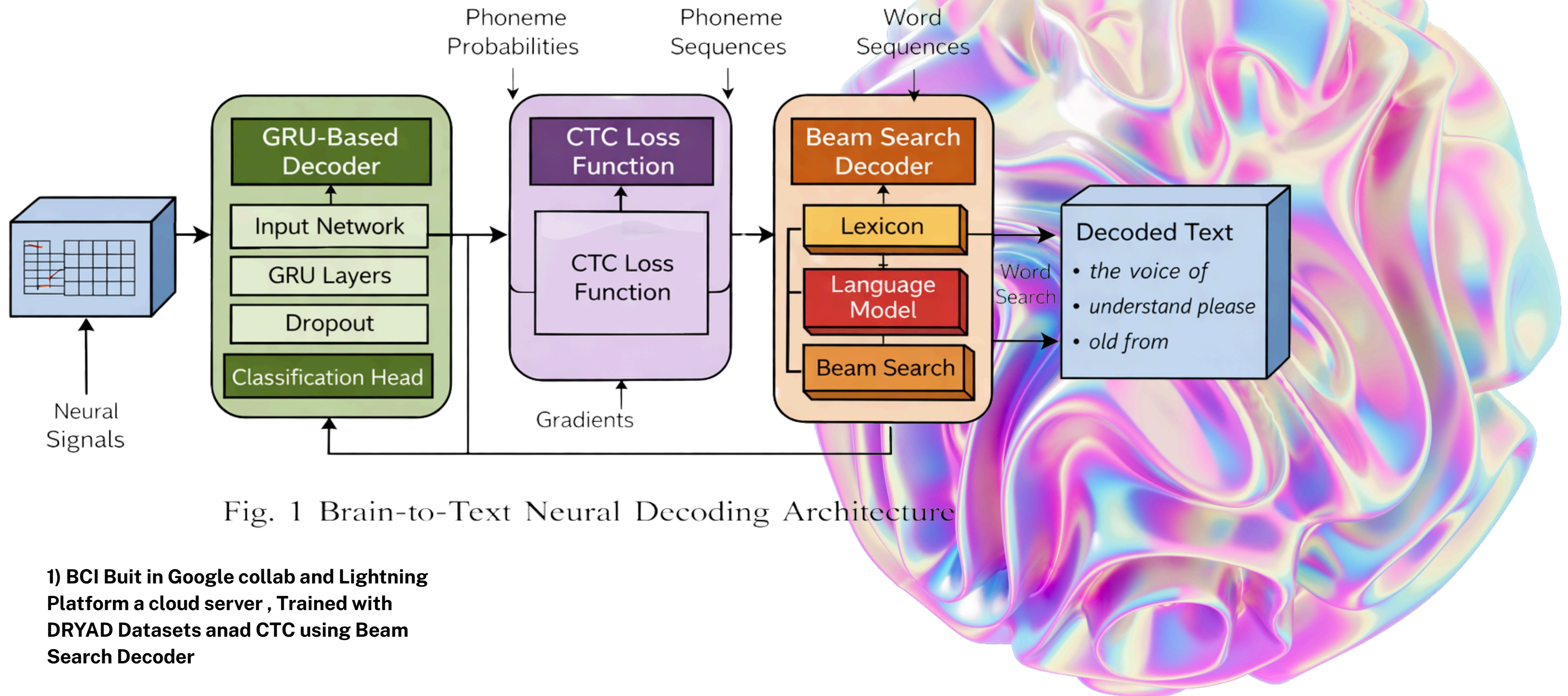


Fig. 1 Brain-to-Text Neural Decoding Architecture

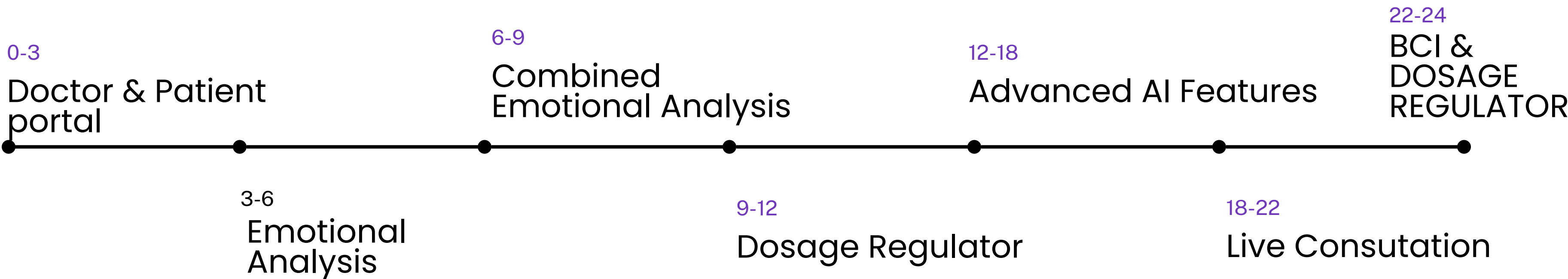
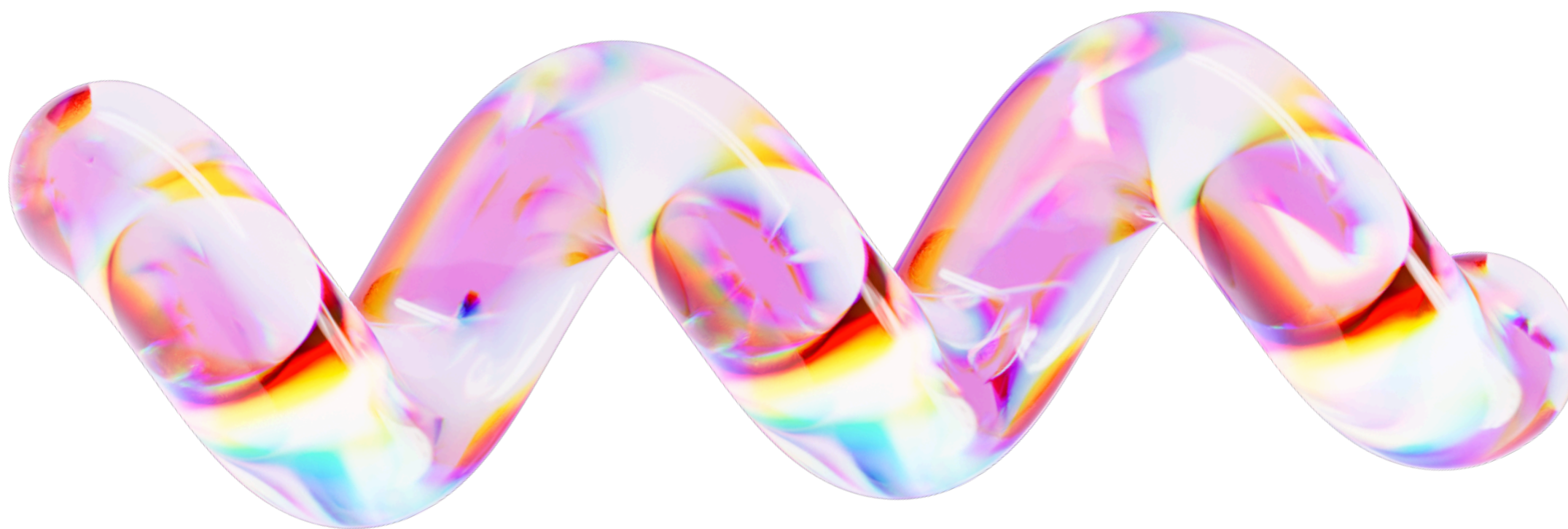
1) BCI Built in Google collab and Lightning Platform a cloud server , Trained with DRYAD Datasets and CTC using Beam Search Decoder

# BRAIN TO TEXT - BCI OUTPUT

| Sample | Ground Truth<br>(Patient)        | Predicted Text<br>(Decoder) | After Language<br>Model       | WER  |
|--------|----------------------------------|-----------------------------|-------------------------------|------|
| 1      | i am feeling pain in<br>my chest | i feeling pain chest        | i am feeling pain in<br>chest | 0.22 |
| 2      | i need help to stand<br>up       | i need help stand           | i need help stand up          | 0.25 |
| 3      | my head is hurting<br>badly      | my head hurt bad            | my head is hurting<br>bad     | 0.24 |
| 4      | i cannot breathe<br>properly     | i cannot breath<br>proper   | i cannot breathe<br>proper    | 0.23 |
| 5      | i feel dizzy when i<br>walk      | i feel dizzy walk           | i feel dizzy when walk        | 0.21 |
| 6      | please give me some<br>water     | please give some<br>water   | please give me water          | 0.26 |
| 7      | my stomach is in<br>pain         | my stomach pain             | my stomach is pain            | 0.24 |
| 8      | i am very tired today            | i am tired today            | i am tired today              | 0.28 |
| 9      | i need medicine for<br>fever     | i need medicine<br>fever    | i need medicine<br>fever      | 0.27 |
| 10     | i cannot sleep at<br>night       | i cannot sleep night        | i cannot sleep night          | 0.29 |



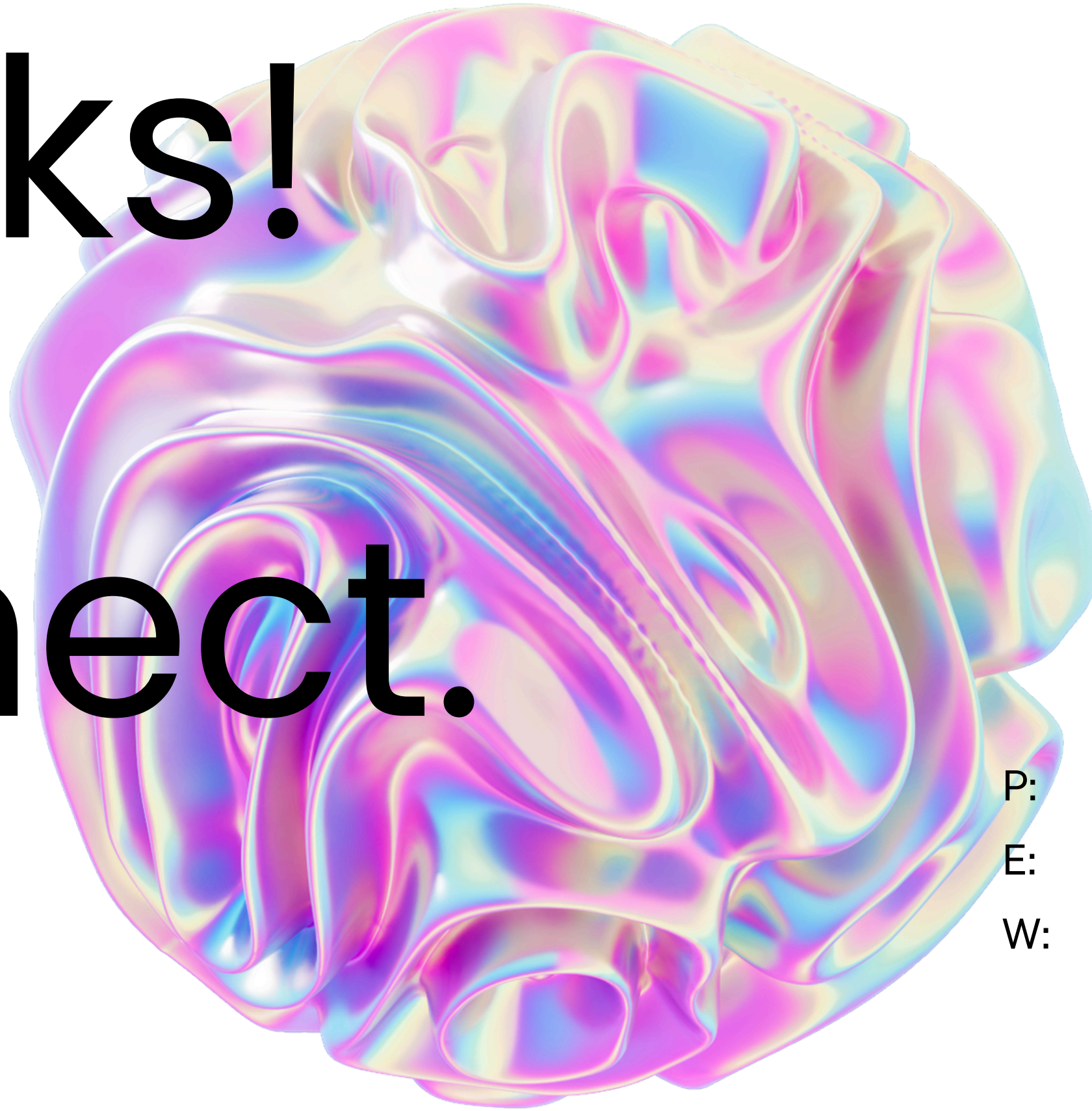
# Milestones & Timeline



# Meet the Innovators



# Thanks! Let's Connect.



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